

ANALYSIS OF UWA COMMUNICATION INDEPENDENT PARAMETERS USING BELLHOP SIMULATOR

*A Project Report Submitted in partial fulfilment of the
requirements for the award of the Degree of*

BACHELOR OF TECHNOLOGY

in

**ELECTRONICS AND COMMUNICATION
ENGINEERING**

Submitted by

JISHNU GANESH 19KD1A0459

Under the Esteemed Guidance of

Mr. Shaik Azeez, M. Tech., (Ph. D)

Assistant Professor, Department of ECE



**Department of Electronics and Communication
Engineering**

**LENDI INSTITUTE OF ENGINEERING AND
TECHNOLOGY
(An Autonomous Institution)**

(Permanently Affiliated to JNTU-GV Vizianagaram, Approved by

AICTE Accredited by NBA, Accredited with NAAC with 'A' Grade)

JONNADA VILLAGE, DENKADA MANDAL,

VIZIANAGARAM DISTRICT, ANDHRA PRADESH, INDIA-535005

2019-2023



LENDI INSTITUTE OF ENGINEERING AND TECHNOLOGY
An Autonomous Institution

**(Permanently Affiliated to JNTU-GV Vizianagaram, Approved by AICTE
Accredited by NBA, Accredited with NAAC with 'A' Grade)**

**Jonnada (Village), Denkada (Mandal), Vizianagaram District,
Andhra Pradesh, India-535005**

CERTIFICATE

This is to certify that the project report entitled “**ANALAYSIS OF UWA COMMUNICATION ON DEPENDENT PARAMETERS USING BELLHOP SIMULATOR**” is the bonafied work of Jishnu Ganesh (19KD1A0459) student of this college, submitted in partial fulfilment of the requirements for the award of degree Bachelor of Technology in Electronics and Communication Engineering during the academic year 2022- 2023.

Head of the Department

Dr. M. Rajan Babu M.Tech., Ph.D., MISTE., FIE

Professor

Department of ECE

Project Supervisor

Mr. Shaik Azeez, M.Tech., (Ph.D),

Assistant Professor

Department of ECE

EXTERNAL EXAMINER

ACKNOWLEDGEMENTS

I consider it as a privilege to thank all the people who helped us for the successful completion of the project work entitled “**Analysis of UWA Communication Using Bellhop Simulator**”.

I would like to express our heartfelt gratitude to our parents for their encouragement and support to achieve and fulfil our dreams.

I take this opportunity to express our deep sense of gratitude to the **Management** of LENDI Institute of Engineering and Technology for providing congenial atmosphere and encouragement, also establishing an R&D Center for the Department of ECE enriched with MATLAB Simulation environment for us.

I sincerely express our wholehearted thanks to **Dr. V. V. Rama Reddy**, Principal, Lendi Institute of Engineering and Technology who has given a lot of support and freedom during our academics.

I profoundly thank **Dr. M. Rajan Babu**, Head of the Department, Electronics and Communication Engineering, for his collaboration, constant support, and positive belief on his students for the successful completion of this project work.

I would like to thank Dean R&D, **Dr. A V Paramkusam**, for providing access to R&D center for successful completion of our project.

I would like to thank our Final year Coordinator **Dr. B. Kiranmai**, Professor, **Dr. T. Anilkumar**, **Sri. B. Hemanth Nag**, **Sri. N. Rajasekhar**, **Sri. S. S. Kiran**, Project Coordinators and the **Project Review Committee Members** for their valuable support and timely help also for providing valuable comments and suggestions to enhance the quality of our project.

I would like to express our sincere thanks to our guide **Mr. Shaik Azeez, Assistant Professor** for his technical guidance, continuous encouragement, and support in carrying out the project work even though he is busy with his hectic schedule. His contagious enthusiasm, positive attitude and an open-door policy for his students greatly helped in this work.

I are thanking the mentor **Mr. Shaik Azeez** of **communication and signal processing Research Group** for providing assistance, resources and constructive suggestions throughout our project work through their workshops in association with MoU organizations and IE(I) & IETE Professional Chapters..

Finally, I would like to thank all the Teaching and Non-Teaching Staff who helped us in the successful completion of this project work. We would also like to thank all of our friends who helped us directly and indirectly for the successful completion of our project work.

JISHNU GANESH 19KD1A0459



LENDI INSTITUTE OF ENGINEERING AND TECHNOLOGY
An Autonomous Institution

**(Permanently Affiliated to JNTU-GV Vizianagaram, Approved by AICTE
Accredited by NBA, Accredited with NAAC with 'A' Grade)**

**Jonnada (Village), Denkada (Mandal), Vizianagaram District,
Andhra Pradesh, India-535005**

INSTITUTE

VISION

Producing globally competent and quality technocrats with human values for the holistic needs of industry and society.

MISSION

- Creating an outstanding infrastructure and platform for enhancement of skills, knowledge and behaviour of students towards employment and higher studies.
- Providing a healthy environment for research, development and entrepreneurship, to meet the expectations of industry and society.
- Transforming the graduates to contribute to the socio-economic development and welfare of the society through value-based education.

	LENDI INSTITUTE OF ENGINEERING AND TECHNOLOGY An Autonomous Institution (Permanently Affiliated to JNTU-GV Vizianagaram, Approved by AICTE Accredited by NBA, Accredited with NAAC with 'A' Grade) Jonnada (Village), Denkada (Mandal), Vizianagaram District, Andhra Pradesh, India-535005
---	---

Department of Electronics and Communication Engineering

VISION

Emerge as a Centre of Eminence in Electronics and Communication Engineering to impart quality education towards competent and skilled engineers.

MISSION

- Offering an inspiring and conducive learning environment to prepare skilled and competent engineers by providing infrastructure, laboratory facilities, and effective teaching-learning process.
- Fostering culture to face complex technological challenges through Internships, Projects and Industry-Institute Interactions in order to enhance employability skills.
- Creating an environment for higher studies and entrepreneurship by way of imparting quality education and promoting research activities.
- Imparting professional behaviour and strong ethical values towards societal issues by encouraging socially relevant activities.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO1: Graduates shall have strong knowledge and technical skills in core and associated fields of Electronics and Communication Engineering to become globally competent engineers and emerging researchers.

PEO2: Graduates shall comprehend latest tools and techniques in the field of Electronics and Communication Engineering to analyze, design and develop novel systems and products for addressing the real time issues.

PEO3: Graduates shall have professional attitude, ethical values, teamwork and good communication skills, to adapt the rapidly changing technologies in Electronics and Communication Engineering through life-long learning.

PROGRAM OUTCOMES (POs)

- 1. Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- 2. Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- 3. Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- 4. Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- 5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
- 6. The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- 7. Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- 8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- 9. Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- 10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- 11. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- 12. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Capable of design, develop, test, verify and implement analog and digital electronics and communication engineering systems and products.

PSO2: Qualify in national and international competitive examinations for successful higher studies and employment.

COURSE OUTCOMES:

S. No	Description
C412.1	Acquire technical knowledge on fundamental aspects in electronics and communication engineering to solve complex engineering problems for real time applications.
C412.2	Identify the work based on past experiences and from literature survey for specific problems in the field of Engineering.
C412.3	Design or develop a software/Hardware model by using modern tools for a specific problem useful for society/environment real time issues by following ethical values.
C412.4	Enhance technical presentation skills, project management abilities to work individually or as a team inculcating leadership qualities.
C412.5	Identify suitable applications leading to enhanced knowledge in project management and lifelong learning.

Abstract

Simulation forms an important part of the development and empirical evaluation of underwater acoustic network (UAN) protocols. The key feature of a credible network simulation model is a realistic channel model. A common approach to simulating realistic underwater acoustic (UWA) channels is by using specialised beam tracing software such as BELLHOP. However, BELLHOP and similar modelling software typically require knowledge of ocean acoustics and a substantial programming effort from UAN protocol designers to integrate it into their research. This paper is a distilled tutorial on UWA channel modeling with a focus on network simulation, providing a trade-off between the flexibility of low level channel modeling via beam tracing and the convenience of automated channel modeling, e.g. via the World Ocean Simulation System (WOSS). The tutorial is accompanied by our MATLAB simulation code that interfaces with BELLHOP to produce channel data for UAN simulations. As part of the tutorial, we describe two methods of incorporating such channel data into network simulations, including a case study for each of them: 1) directly importing the data as a look-up table, 2) using the data to create a statistical channel model. The primary aim of this paper is to provide a useful learning resource and modeling tool for UAN protocol researchers. Initial insights into the UAN protocol design and performance provided by the statistical channel modeling approach presented in this paper demonstrate its potential as a powerful modeling tool for future UAN research.

Keywords: Acoustic communication, outage criterion, throughput, RLPC, OFDM, sparse channel estimation, Adaptive algorithm.

Software/Equipment/Tools Requirement: Bellhop simulator, MATLAB.

Relevance to POs & PSOs: PO1, PO2, PO3, PO4, PO11, PO12 and PSO1 are attained.

COURSE OUTCOMES VS POs MAPPING:

(DETAILED: HIGH:3; MEDIUM:2; LOW:1)

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
C412.1	3	3	3	2	3	1	1	1	1	3	2	2	3	1
C412.2	3	3	3	2	3	1	1	1	1	3	2	2	3	1
C412.3	3	3	3	2	3	1	1	1	1	3	2	2	3	1
C412.4	3	3	3	2	3	1	1	1	1	3	2	2	3	1
C412.5	3	3	3	2	3	1	1	1	1	3	2	2	3	1
C412*	3	3	3	2	3	1	1	1	1	3	2	2	3	1

Table of Contents

1	Introduction	1
	Background	2
	1.1 Applications of Underwater wireless acoustic networks	3
	1.2 Overview of underwater wireless communication	3
	1.3 Underwater acoustic communication	5
	1.4 Data range for underwater acoustic communication	6
	1.5 UWAC Fundamentals	6
	1.5.1 Sound	6
	1.5.1.1 Acoustic Pressure	6
	1.5.1.2 Acoustic Impedance	6
	1.5.1.3 Acoustic Intensity	7
	1.5.2 Acoustic Waves	7
	1.5.3 Acoustic Source level	7
	1.6 Underwater channel characteristics	8
	1.6.1 Spreading Loss	8
	1.6.2 Absorption Loss	9
	1.6.3 Path Loss	11
	1.6.4 Channel Bandwidth	11
	1.6.5 Noise	12
	1.6.6 Research on UWAC	14
2	Literature Review	15
	2.1 Literature Analysis	16
3	Proposal Work	20
	3.1 Introduction	21
	3.2 Methodology	21
	3.3 Transmission Range	21
	3.4 Transmission Loss	22
	3.5 Frequency	23
	3.6 Depth	24

3.7 Environmental conditions	25
3.8 Bellhop	26
4 Bellhop simulator	27
4.1 Need of Bellhop	29
4.2 Bellhop Implementation	30
4.3 Bellhop Acoustic Toolbox	31
4.3.1 Installing and Launching AcTUP	31
4.3.2 Installing Files	32
4.3.3 Setting Up AcTUP	32
4.3.4 Running AcTUP	33
4.3.5 Summarizing Bellhop Simulator	33
4.3.6 Main Menu	34
4.3.7 Configure Environment & Propagation Models	34
4.3.8 Select Active Code	34
4.3.9 Run Current Model for Active Propagation Code	35
4.3.10 Run Current Model for All Propagation Code	35
4.3.11 Run Model(s) from File(s)	35
4.3.12 View Log or Environment File	35
4.3.13 Environmental & Propagation Definitions	35
4.3.14 Load Default Run Definition	36
4.3.15 Load Run Definition	36
4.3.16 Save Run Definition	36
4.3.17 Edit Environment	36
4.3.18 Edit Code-Dependent Propagation Parameters	36
4.3.19 Plotting Tools	37
5 Results and conclusion	39
5.1 Results	40
5.2 Conclusion	42
5.3 Outcomes	43
5.4 Future scope	43
5.5 References	44

List of Figures

Figure No.	Figure Caption	Page No.
Figure 1.	Overview of underwater wireless communication	4
Figure 2.	Overview of underwater acoustic communication	5
Figure 3.	Transmission loss l	23
Figure 4.	Bellhop Structure	24
Figure 5.	Delay vs Depth	40
Figure 6.	Delay vs Range	40
Figure 7.	Horizontal wavenumber vs Magnitude of greens function	41
Figure 8.	Real vs Imaginary	41
Figure 9.	Range vs Depth	42

List of Tables

Table No.	Caption of the Table	Page No.
Table 1.	Applications of Wireless Communications	3
Table 2.	Data Range for underwater acoustic communication	6
Table 3.	Bandwidth Limitatiions	12
Table 4.	Ascii format	37

CHAPTER 1

INTRODUCTION

Chapter-1

Introduction

Background

Around 70% of the surface of earth is covered with water. It contains large amount of energy and natural resources and hence for exploring these resources, development of effective underwater wireless communication system became very important. Due to number of applications like scientific marine exploring, pollution controlling, voice and data communications between divers, mine reconnaissance, study of disasters, military projects, etc. The field of underwater wireless communication is rapidly increasing. Also, shipbuilding and offshore engineering is taking lot of interest in it. So, obviously it is one of the important

Underwater wireless communication is an enabling technology for oceanography and its applications. It consists of number of vehicles with attached sensors for monitoring the given area. But, development of underwater wireless communication system is not that much easy as it is totally different from terrestrial radio environment both in terms of energy, cost and channel propagation.

Major challenges encountered in the design of underwater acoustic networks are as follows

- The available bandwidth is very limited which causes reduction in transmission range
- The underwater channel is mainly impaired due to multipath and fading
- Propagation delay in underwater wireless communication is five times higher than RF terrestrial channel
- Underwater sensors are very costly. And very high protection is required for these sensors. Also most of the times they fail due to corrosion
- Battery power is very limited and it can't be charged as solar energy can't be exploited.

Out of EM, MI and acoustic waves most of the applications are done by using acoustics as they are proven to be excellent in underwater compare to others as they provide wide range up to 10 km with very small antenna size nearly about 0.1m. Also,

power loss in underwater acoustics are very small compare to EM and optical waves in sea environment.

The basic problems in front of underwater acoustic communication are limited sound velocity, absorption of a sound in water, multipath propagation, ambient noise, Doppler's spread, etc.

1.1 Applications of Underwater wireless acoustic networks

The applications of underwater wireless communication are given in terms of environmental, marine, scientific and military system.

Environm ental	Marine	Scientific	Military
Pollution	Finding natural resources	Oceanogra phy	Bottom imaging
Finding oil and gas source s	To study marine life	Geo sciences	Detection of underwater objects
Protection of ships	Finding resources in deepsea	Marine biology	Threat detection
Weather study	Pollution controlling	Seismic study	AUV controlling

Table 1. Applications of wireless communication

1.2 Overview of underwater wireless communication

Underwater wireless acoustic communication system is a two way communication system in which digital data is converted into special underwater sound signals. Then these signals are received by another transreceiver and then converted back to digital data. This communication is done in multipath propagation of acoustic/sound signal.

The underwater wireless communication is come in to picture in 1940 when Leonardo Da Vinci said, “If you cause your ship to stop and place the head of long tube into water and place extremity to your ear you will hear sound of ships standing at a long distance from you”. In 1912, after incidence of Titanic, systems for detection of icebergs are developed which arealso used in World War.

Underwater wireless communication consists of a transreceiver with antennas and

sensors attached to them. As discussed above underwater sensors are coated very carefully to protect them from corrosion. We can transmit EM or acoustic wave but acoustics is a main choice as it gives better result and better range.

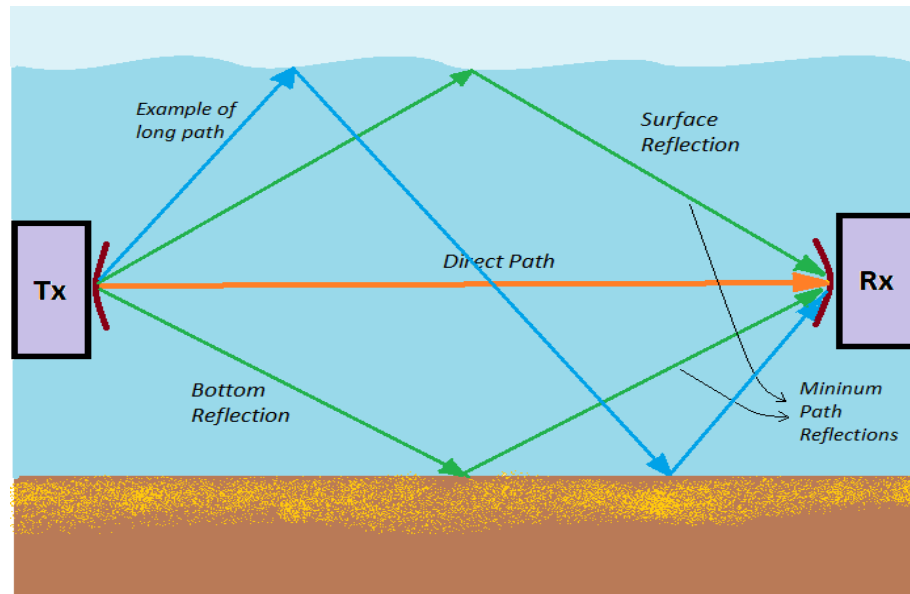


Figure 1. Overview of Underwater wireless communication

Due to attenuation, the transmitted signal transmits in a form of multipath. At receiver every multipath signal must be recovered to get original signal back. The reflection of these multipath signals is of two types a surface reflection and bottom reflection. Surface reflection depends upon surface properties while bottom reflection depends on bottom type and sand properties.

As some multipath signals become long path signals and some signals become short path signals hence propagation delay for every multipath signal is different which increases overall propagation delay of a transmission signal.

If we use acoustic waves for transmission then it mainly affects due to thermal noise and converts into heat which reduces system efficiency hence, usually 20 KHz to 100 KHz frequency band is selected for acoustics.

Underwater wireless communications play an important role in marine activities such as environmental monitoring, underwater exploration, and scientific data collection. Underwater wireless communications still remain quite challenging, due to the unique and harsh conditions that characterize underwater channels.

These conditions include for example severe attenuation, multipath dispersion, and limited resource utilization. Nonetheless advanced communication techniques using acoustic, electromagnetic and/or optical waves have emerged to tackle fundamental and practical challenges of underwater wireless communications.

1.3 Underwater acoustic communication

Underwater acoustic communication is a technique of sending and receiving messages below water. There are several ways of employing such communication but the most common is by using hydrophones. Underwater communication is difficult due to factors such as multi-path propagation, time variations of the channel, small available bandwidth and strong signal attenuation, especially over long ranges. Compared to terrestrial communication, underwater communication has low data rates because it uses acoustic waves instead of electromagnetic waves.

At the beginning of the 20th century some ships communicated by underwater bells as well as using the system for navigation. Submarine signals were at the time competitive with the primitive maritime radionavigation. The later Fessenden oscillator allowed communication with submarines.

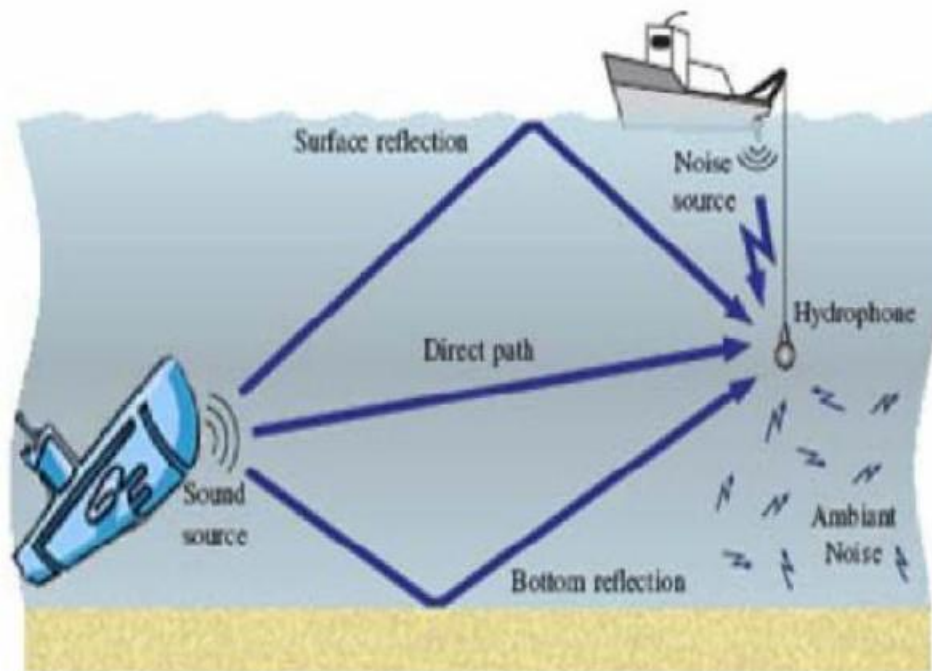


Figure 2. Overview of Underwater acoustic communication

An underwater transmitter directs a sonar signal to the water's surface, causing tiny vibrations that correspond to the 1s and 0s transmitted. Above the surface, a highly sensitive receiver reads these minute disturbances and decodes the sonar signal.

1.4 Data Range for underwater acoustic communication

Range	Frequency
1000 Km	Less than 1 KHz
Up to 100 Km	2 to 5 KHz
Up to 10 Km	Nearly 10 KHz
Up to 1 Km	20 to 50 KHz

Table 2. Data Range for underwater wireless acoustic communication

1.5 UWAC Fundamentals

As in the electrical area, there are also in acoustic important physical quantities that must be described to give the necessary background associated with this field of knowledge.

1.5.1 Sound

Sound is produced when an object vibrates and transmits his motion to the surrounding physical medium. This results in propagation of vibrations where the particles in the medium oscillate in the same direction of the propagation, so we have what is called a longitudinal wave.

1.5.1.1 Acoustic Pressure

Given a plane wave, acoustic Pressure (P), with unit Pa or N/m, is defined by the following equation is $P = P_0 \cos(\omega t - kx) = P_0 \cos(\omega t - 2\pi f \Sigma, \Sigma = V/(2\pi f))$.

Where represents the fluid density, c is the velocity of the sound wave propagation and is the particle velocity. The variable v is equivalent to $2\pi f$. This quantity analogous to the potential difference in electrical circuits. The quantity is called specific impedance and has the same role as the intrinsic impedance defined for a transverse electromagnetic.

1.5.1.2 Acoustic impedance

The acoustic impedance is given by $Z = P/U$. Where U is the acoustic volume flow. This equation is analogous to Ohm's law and Z is a function of frequency, with real and imaginary components.

1.5.1.3 Acoustic Intensity

The acoustic intensity (watt/m²) is the energy per second that crosses the unit area. For a plane wave it is given by $I = P \cdot v$.

Normally, a reference intensity I is defined for each medium under certain circumstances. For example, the underwater reference intensity is the one produced by a plane wave with root mean square pressure of 1 Pa.

1.5.2 Acoustic Waves

As an example, the wave equation of a baffled piston projector, a circular kind of projector, in spherical coordinates (r, θ) , is given by

$$p(r, \theta, t) = j p_c (Q k / 2 \pi r) e^{j(\omega t - k r)} 2 J_1(k a \sin \theta) / k a \sin \theta$$

Where the first term represents a spherical wave generated by what is called an ideal omnidirectional source. The second term, which depends of θ and the Bessel function of the first kind, represents the directivity of the actual source. It gives us an idea on how the source concentrates more energy on certain special directions. This is very

analogous to the antenna field theory and we can, in a similar way, define the near field and the far field. This second one is of interest from the acoustic wave propagation point of view, as it can be approximated by a one-dimension front wave and it can be treated as a ray, or a sum of rays, in different directions, when the distance is longer than the wavelength.

1.5.3 Acoustic source level

The source level quantity associated with a projector, SL projector, is commonly defined in terms of the sound pressure level at a well-defined distance of 1m from its acoustic center. The source intensity at this reference point is

$$I = (P_{tx}/Area)(W/m^2)$$

1.6 Underwater Channel Characteristics

Physical and chemical properties of sea water affect sound propagation. Due to spreading and absorption, an underwater acoustic signal will suffer attenuation. Furthermore, depending on channel geometry, multi-path fading may occur and produce significant inter-symbol-interference (ISI) at the receiver hydrophone. For calculations of the Signal-to-Noise ratio (SNR) or Bit-Error-Rate (BER) estimation, it is then crucial to understand and establish a good channel model.

1.6.1 Spreading Loss

Spreading loss is due to the ever-increasing area covered by the same amount of the sound signal energy, as a wave front moves outward from the source. It is given by

$$PL_{spreading}(r) = k * 10 \log(r) (dB)$$

When the medium in which signal transmission occurs is unbounded, the spreading factor is $k=2$, meaning that the source intensity decreases with the square of the distance. In the case of bounded spreading this factor takes different values. For example, $k=1$ for a cylindrical boundary

In 1967, Urick suggested that spherical spreading was a rare occurrence in the ocean

Department of ECE, Lendi Institute of Engineering & Technology(Autonomous) Page 8

but recognized that at short ranges it may occur. As Autonomous Underwater Vehicles (AUV) swarm operations and Underwater Sensor Networks (USN) will typically be short range applications.

It is likely that spherical spreading will need to be considered the primary factor leading to signal attenuation, in those cases. Spreading loss has a logarithmic relationship with range and its impact on the signal is most significant at very short range, which is up to approximately 50m .

At these shorter ranges, spreading loss plays a proportionally larger part when compared with the absorption loss, which is the topic of the following subsection.

1.6.2 Absorption Loss

The absorption loss represents the energy loss in the form of heat, due to the viscous friction and ionic relaxation that occurs as the sound wave propagates outwards underwater. This loss may be given by the following generic equation is

$$PL_{\text{absorption}}(r,f)=10\log(\alpha(f))*r(\text{dB})$$

Where α is the absorption coefficient appears on frequency for a given frequency values absorption varies linearly with range in kilometre's. More specifically, the absorption of sound in sea water is caused by three dominant effects, namely viscosity, ionic relaxation of boric acid and magnesium sulphate (MgSO_4) molecules and the relaxation time.

The effect of viscosity is significant at the high frequencies above 100 kHz. whereas the ionic re-laxation effects of magnesium affect the mid frequency range from 10 kHz up to 100 kHz, and boric acid effects dominate at low frequencies, which go up to few kHz. In general, the absorption coefficient, α , increases with increasing frequency and decreases as depth increases. Its value is significantly higher for sea water when compared to fresh water, due predominately to the ionic relaxation factor. Extensive measurements of absorption losses over the last half century led to several empirical formulae which take into account frequency, salinity, temperature, pH, depth and speed of sound.

A popular version is Thorp's expression, which is based on his initial investigations in the 60's and has then been converted into metric units considered in equation 8.

It is valid for frequencies from 100 Hz to 1 MHz and is based on sea water with salinity of 35 ppt (parts per thousand), pH of 8, and temperature of 4°C and depth of 0m (atmospheric pressure) which is assumed but not stated by Thorp.

$$\alpha(f)=(0.11f^2/1+f^2)+(44f^2/4100f^2)+275 \times 10^{-1} f^2 + 0.0033(\text{dB/km})$$

Fisher and Simmons (1977) and others, Francois & Garrison (1982), have proposed other variations for the absorption coefficient. In particular, Fisher and Simmons in the late 70's found the effect on the absorption coefficient associated with the relaxation of boric acid.

They provided the following more detailed form for α , which varies with frequency, pressure (depth) and temperature (also valid for 100Hz to MHz with salinity 35 ppt and acidity 8 pH).

$$\alpha(f,d,t)=(A_1f_1^2/f_1^2+f^2)+(A_2P_2f_2^2/f_1^2+f^2)+A_3P_3f^2(\text{dB/km})$$

In this equation d is depth in meters and t is temperature in °C. The 'A' coefficients represent the effects of temperature, while the P 'coefficients represent ocean depth (pressure) and 2 represent the relaxation frequencies of Boric acid and (MgSO) molecules. These terms developed by Fisher and Simmons were also considered more recently by Domingo Figure 5 shows the absorption coefficients in dB/km vs signal frequency for both Thorp, and Fisher and Simmons equations.

In general, α increases with increasing frequency for any fixed temperature and depth. Up until around 80 kHz, a change in temperature has a more significant effect on α than depth, but above these frequencies' depth becomes the dominant factor in any case, Thorp's approximation is quite close to Fisher and Simmons and is clearly more conservative at the frequencies shown. It has been shown that at frequencies above 300 kHz Thorp's model predicts lower losses, as it does not take into account the relaxation frequencies found by Fisher and Simmons.

If depth and frequency are fixed and temperature varied from 0 to 27°C. there is a decrease in α of approximately 4dB/km for frequencies in the range of 30 to 60 kHz, which correlates to work presented by Urlick.

If we consider where AUV swarms are most likely to operate, in the 'mixed surface layer, where temperature varies considerably due to latitude, but with an average value of 17°C temperature variations may be an important factor. It should be noted that when operating at lower temperatures α will have a higher value and, so, using 0°C will be a conservative alternative. At shorter ranges, the significance of α is expected to be less than spreading loss due to the linear relationship of absorption loss with range. As mentioned, depth (pressure) has less of an effect on α than temperature at lower frequencies.

Domingo investigated the effect of depth (pressure) on absorption and confirmed that for frequencies below 100 kHz there is less change in α . Urick also studied this variation and proposed the following formula:

$$\alpha_d = \alpha \cdot 10^{-3} d(1 - 5.9 \cdot 10^{-6}) \text{ (dB)m}$$

Where d is depth in meters, but he also suggested as an approximation to take a 2% decrease for every 300 m depth change. Thus, depth (pressure) variations are not expected to play a significant role in short range AUV swarm operations, especially those that use a 2D horizontal topology as described.

1.6.3 Pathloss

Total path loss, or just path loss, is the combined contribution of both the spreading and absorption losses. In the year of 1967, Urick established that the formula below, for spreading plus absorption, yields areas in able agreement with long term observations.

$$\text{Path Loss}(r, f, d, t) = k \cdot 10 \log(r) = \alpha(f, d, t) r \cdot 10^{-3} \text{ (dB)}$$

1.6.4 Channel Bandwidth

The attenuation or path loss that occurs in an underwater acoustic channel for a signal of frequency f and over a distance d may also be given by the following equation:

$$A(l, f) = dk \cdot a(f)^d$$

Where A is the spreading factor (typically the value of 1, 5 is used), and α is the absorption coefficient Using Thorp's formula, which gives of more precisely than other models.

For frequencies above about 1001, as we have seen, the absorption coefficient can be written empirically, the main limitative factor that cuts the maximal desired range of frequencies usable in an underwater system is the absorption loss coefficient.

Raising the frequency, leads to an increase of the path loss due to the increase in the absorption coefficient, turning the frequencies above a certain threshold unusable for acoustic underwater communications. This bandwidth limitations are:

	Range (km)	Bandwidth (kHz)
Very Long	1000	< 1
Long	10 - 100	2 - 5
Medium	1 - 10	10
Short	0,1 - 1	20 - 50
Very Short	< 0,1	> 100

Table 3. Bandwidth limitations

1.6.5 Noise

For the underwater channel four noise components are considered. There is the thermal noise (N_a), there is noise due to the movement of waves (N_w), making the water medium unstable and varying the that can be taken as additive white Gaussian noise that is always present in communications systems.

Then, static properties of the channel. Another type of noise is water movement caused by ship traffic (N_s) and, finally.

There is also noise due to natural causes (NA)), like the turbulence caused by storms or during the rain events and the presence of bubbles.

The total noise can be obtained adding of all these contributions as shown by the equations.

Turbulence Noise

$$N_t = 10((17 - 30 \log(f))/10) N_{toff}$$

Shipping Noise

$$N_s = 10((40 + (20(s - 0.5)) + (26 \log_{10}(f)) - (60 \log_{10}(f + 0.03))))/10 \\ *(N_{soff})$$

Wave Noise

$$N_w = 10(50 + 75(w^{1/2}) + (20 \log_{10}(f)) - (40 \log(f + 0.4)))/10 \\ *(N_{woff})$$

Thermal Noise

$$N_{th} = 10((-15 + 20 \log_{10}(f))/10) *(N_{thoff})$$

Total Noise

$$N = N_t + N_s + N_w + N_{th}$$

1.6.6 Research on UWAC

Several research groups are actively working in the UWAC area. At the Massachusetts Institute of Technology, a group headed by Prof. Milaca Stojakovic, has been working in this field, investigating the use of MIMO and OFDM technique. Also, a group from Bremen University of Applied Sciences, has been working in UWAC with emphasis on modelling and simulation of the communication systems. As another example, the Broadband Wireless Network in Lab from the Georgia Institute of Technology, has been working in the topic of underwater acoustic sensor networks(UW-ASN).

CHAPTER 2

LITERATURE REVIEW

CHAPTER-2

LITERATURE REVIEW

2.1 Literature Analysis

Underwater acoustic communication (UAC) is a specialized field that involves the transmission of information through acoustic waves in water. UAC is widely used in applications such as underwater navigation, oceanographic data collection, environmental monitoring, and marine biology research. The following literature review provides an overview of some of the key research on underwater acoustic communication.

1. "Modeling and Analysis of Underwater Acoustic Communication Channels with Bellhop Simulator" by Shibo Zhang and Lijun Sun. This paper presents a comprehensive analysis of UWA communication independent parameters, including frequency response, distance attenuation, water temperature, water salinity, and source level, using Bellhop simulator. The authors also compare the simulation results with experimental data to validate the accuracy of the simulation.
2. "Simulation of Underwater Acoustic Communication Channels using Bellhop" by Ming Li, Liang Liang, and Baoling Han. This paper presents a detailed analysis of the frequency response and distance attenuation of UWA communication channels using Bellhop simulator. The authors also evaluate the performance of different communication techniques, such as orthogonal frequency division multiplexing (OFDM), in different water environments.
3. "Analysis of Underwater Acoustic Communication Channels using Bellhop" by Zhongshan Zhang, Guangzhi Zhang, and Guangjie Han. This paper presents a comprehensive analysis of the effects of water temperature, water salinity, and source level on UWA communication using Bellhop simulator. The authors also propose a novel algorithm for optimizing the transmission parameters based on the simulation results.
4. "Bellhop-based Simulation of Underwater Acoustic Communication Channels" by Kostas Peppas and Nikolaos Frangiadakis. This paper presents an analysis of

UWAcommunication independent parameters, including frequency response, distance attenuation, water temperature, and water salinity, using Bellhop simulator.

The authors also evaluate the performance of different communication techniques, such as phase modulation and amplitude modulation, in different water environments.

5. The paper "Performance Analysis of Underwater Acoustic Communication System using Bellhop Simulator" by R. Rizk and M. El-Hajjar (2021) discusses the use of the bellhop simulator to analyze the effect of different independent parameters on the performance of underwater acoustic communication systems. The authors specifically focus on the effect of water temperature, water salinity, and source level on the system's performance.
6. In the paper "Underwater Acoustic Channel Characterization and Modeling for Communication System Design and Performance Analysis" by S. Soni, et al. (2019), the authors discuss how the bellhop simulator can be used to analyze the effect of different independent parameters on the performance of underwater acoustic communication systems. The paper specifically focuses on the analysis of frequency response and signal strength attenuation due to distance and water temperature.
7. In the paper "Underwater Acoustic Channel Characterization and Modeling for Communication System Design and Performance Analysis" by S. Soni, et al. (2019), the authors discuss how the bellhop simulator can be used to analyze the effect of different independent parameters on the performance of underwater acoustic communication systems. The paper specifically focuses on the analysis of frequency response and signal strength attenuation due to distance and water temperature.
8. The paper "Modeling Underwater Acoustic Channels for Improved Communication Systems" by T. Abreu, et al. (2018) discusses the use of the bellhop simulator to analyze the performance of underwater acoustic communication systems under different environmental conditions. The authors specifically focus on the effect of distance attenuation, water temperature, and water salinity on the system's performance.
9. In the paper "Simulation and Experimental Study of Underwater Acoustic Communication Systems" by X. Dong, et al. (2019), the authors discuss how the bellhop simulator can be used to analyze the effect of different independent parameters on the performance of underwater acoustic communication systems.

The paper specifically focuses on the analysis of frequency response and signal strength attenuation due to distance and water temperature.

10. "Evaluation of Underwater Acoustic Communication Performance using Bellhop Simulation Model" by H. Y. Cui, Y. B. Chen, and J. L. Gao. In this paper, the authors use the Bellhop simulator to evaluate the performance of an underwater acoustic communication system under different frequency ranges, distances, and water conditions. The study also evaluates the effect of different source levels on the system performance.

11. "Analysis of Underwater Acoustic Communication System Performance in Different Water Conditions" by W. Li, L. Li, and H. Wang. In this paper, the authors use the Bellhop simulator to analyze the performance of an underwater acoustic communication system under different water temperatures and salinity levels. The study evaluates the effect of these parameters on the signal-to-noise ratio (SNR) and the bit error rate (BER) of the communication system.

12. "Optimization of Underwater Acoustic Communication System Design using Bellhop Simulation Model" by X. Yu, W. Zhang, and X. Liu. In this paper, the authors use the Bellhop simulator to optimize the design of an underwater acoustic communication system. The study evaluates the effect of different transmitter and receiver parameters on the system performance, such as source level, receiver sensitivity, and transducer directionality.

13. "Analysis of the Effect of Distance Attenuation on Underwater Acoustic Communication System Performance" by J. Zhang, X. Gao, and J. Sun. In this paper, the authors use the Bellhop simulator to analyze the effect of distance attenuation on the performance of an underwater acoustic communication system. The study evaluates the effect of different distances between the transmitter and receiver on the system performance, such as SNR and BER.

14. "Simulation Analysis of Underwater Acoustic Communication Performance in Shallow Water" by Y. Ma, X. Li, and X. Huang. In this paper, the authors use the Bellhop simulator to evaluate the performance of an underwater acoustic communication system in shallow water environments. The study evaluates the effect of different water depths, temperatures, and salinity levels on the system performance, as well as the effect of ambient noise on the system performance.

15. "Underwater Acoustics: Analysis, Design, and Performance of Sonar" by Richard P. Hodges: This book provides a comprehensive introduction to the field of underwater acoustics, covering topics such as the physical properties of sound in water, sound propagation, and the design of sonar systems.
16. "Underwater Acoustic Modelling and Simulation" by Paul C. Etter: This book provides a detailed overview of the mathematical and computational methods used in underwater acoustic modelling and simulation, including topics such as ray tracing, normal mode theory, and finite element methods.
17. "Underwater Acoustic Networking Techniques" by Wenbo He and K. J. Ray Liu: This book provides an overview of the networking techniques used in underwater communication systems, including topics such as acoustic propagation modeling, channel estimation, and multiple access techniques.

CHAPTER 3

PROPOSAL WORK

CHAPTER-3

PROPOSAL WORK

3.1 INTRODUCTION

Underwater acoustic communication is an essential part of many applications such as underwater sensing, oceanography, and defense operations. The performance of underwater acoustic communication systems is highly dependent on various parameters such as transmission range, frequency, depth, and environmental conditions. In this project, we aim to analyze the impact of dependent parameters on the performance of underwater acoustic communication systems.

3.2 METHODOLOGY

We will use the Bellhop simulator to simulate the underwater acoustic communication system. The Bellhop simulator is a widely used simulation tool for underwater acoustic communication. It is a ray-tracing-based model that uses the principles of acoustics to simulate the propagation of sound waves in underwater environments. We will perform simulations by varying the dependent parameters such as transmission range, frequency, depth, and environmental conditions.

3.3 TRANSMISSION RANGE

Transmission range is one of the most critical parameters in underwater acoustic (UWA) communication systems. It refers to the distance that an acoustic signal can travel between the transmitter and receiver without significant loss or degradation of the signal. The transmission range is affected by various factors such as the frequency of the signal, the depth of the water, and the properties of the medium through which the signal travels. In general, the transmission range of UWA communication systems is limited due to the high attenuation of sound waves in water. The attenuation is caused by various factors such as absorption, scattering, and reflection of sound waves as they propagate through water. The attenuation increases with increasing frequency, depth, and distance traveled, which reduces the transmission range.

To overcome the limitations of transmission range in UWA communication systems, various techniques such as signal processing, multi-path propagation, and adaptive modulation have been proposed. These techniques aim to improve the signal-to-noise ratio (SNR) of the system and extend the transmission range. Transmission range is a critical parameter in UWA communication systems, and its optimization is crucial for achieving reliable communication in underwater environments.

3.4 TRANSMISSION LOSS

Transmission loss is a phenomenon that occurs when acoustic signals are transmitted through water in underwater acoustic (UWA) communication systems. It refers to the decrease in signal power or intensity as the signal travels through water. The transmission loss is caused by various factors such as absorption, scattering, and reflection of sound waves as they propagate through water.

The attenuation of sound waves due to absorption is one of the primary causes of transmission loss in UWA communication systems. The attenuation is caused by the absorption of sound energy by the water itself and the absorption by various substances present in the water such as dissolved gases, plankton, and suspended particles. The absorption of sound energy increases with increasing frequency and distance traveled, which reduces the signal power and results in transmission loss.

Scattering and reflection of sound waves by objects in the water also contribute to transmission loss. When sound waves encounter an object, they can be scattered or reflected in different directions, resulting in a decrease in the signal power.

To minimize transmission loss in UWA communication systems, various techniques such as adaptive modulation, error-correction coding, and signal processing are used. These techniques aim to increase the signal-to-noise ratio (SNR) of the system, reduce the impact of transmission loss on the signal quality, and improve the reliability of communication.

Transmission loss is a significant factor that affects the performance of UWA communication systems, and its mitigation is crucial for achieving reliable communication in underwater environments.

Transmission loss is a significant factor in underwater acoustic (UWA) communication systems, as it determines the amount of signal attenuation as it propagates through the water. Transmission loss is a function of several parameters, including frequency, distance, depth, and the properties of the medium, such as temperature, salinity, and pressure.

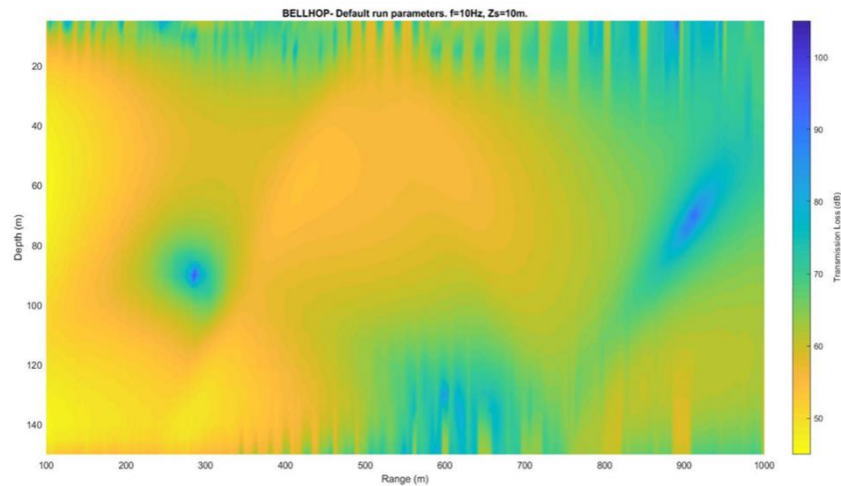


Figure 3. Transmission loss

3.5 FREQUENCY

Frequency is a crucial parameter in underwater acoustic (UWA) communication systems as it affects the signal propagation and transmission characteristics in water. The choice of frequency depends on several factors, such as the desired range of communication, the type of application, and the environmental conditions in which the system will operate.

Low-frequency signals, such as those in the range of 10-100 kHz, can propagate long distances in water, making them suitable for long-range communication. However, low-frequency signals are more susceptible to interference and noise, and the data rate they can support is relatively low.

High-frequency signals, such as those in the range of 10-100 kHz, can support higher data rates, but they are more susceptible to attenuation and signal degradation, which limits their range.

In general, the choice of frequency depends on the specific requirements of the application. For example, in oceanographic applications, lower frequency signals may be preferred to enable long-range communication.

While in defense applications, higher frequency signals may be required for higher data rates and increased security.

Moreover, the frequency of the signal can also affect the performance of other parameters, such as transmission loss, signal-to-noise ratio (SNR), and multipath propagation. Therefore, it is essential to carefully consider the choice of frequency in UWA communication system design, taking into account the application requirements and environmental conditions.

3.6 DEPTH

Depth is an important factor to consider in underwater acoustic (UWA) communication systems as it affects the propagation characteristics of acoustic signals in water. As depth increases, the water pressure and density increase, leading to changes in the speed of sound, attenuation, and scattering of acoustic signals.

In general, as depth increases, the speed of sound in water increases, and the attenuation of the signal due to absorption and scattering increases. The increase in attenuation reduces the signal strength, which reduces the maximum range of communication. At the same time, the increase in speed of sound also changes the direction of propagation of the signal, which can lead to signal distortion and multipath propagation.

To mitigate the effects of depth on UWA communication systems, several techniques are used. One of the common techniques is to use signal processing algorithms, such as equalization and echo cancellation, to mitigate the effects of multipath propagation and signal distortion. Another technique is to use frequency-dependent transmission schemes, where the frequency of the signal is chosen based on the depth of the water to achieve better propagation characteristics.

Depth is an important parameter to consider in UWA communication system design, and its effects on signal propagation should be carefully analyzed and accounted for in the system design.

3.7 ENVIRONMENTAL CONDITIONS

Environmental conditions are an important factor to consider in underwater acoustic (UWA) communication systems, as they can significantly affect the propagation characteristics of acoustic signals in water.

The environmental conditions that can affect UWA communication systems include water temperature, salinity, pressure, and turbidity.

Water temperature affects the speed of sound in water, which in turn affects the propagation speed of acoustic signals. As water temperature increases the speed of sound in water also increases, which affects the direction of propagation and can lead to signal distortion and multipath propagation.

Salinity also affects the speed of sound in water, with higher salinity leading to a higher speed of sound. The salinity of water can vary depending on the location and depth, which can affect the propagation characteristics of acoustic signals.

Water pressure increases with depth and can affect the attenuation and scattering of acoustic signals. As water pressure increases, the attenuation of the signal due to absorption and scattering also increases, reducing the signal strength and limiting the range of communication.

Turbidity, or the amount of suspended particles in water, can also affect UWA communication systems by scattering and attenuating the acoustic signal. High turbidity can reduce the signal strength and limit the range of communication.

To overcome the effects of environmental conditions on UWA communication systems, signal processing techniques, and frequency-dependent transmission schemes are used. These techniques adjust the signal characteristics and frequency to optimize the communication performance based on the environmental conditions.

Overall, environmental conditions are a critical factor in UWA communication system design, and their effects on signal propagation should be carefully analyzed and accounted for in the system design.

3.8 BELLHOP

Bellhop simulator is a widely used tool in underwater acoustic (UWA) communication systems for simulating the propagation of acoustic signals in water. It is a ray-tracing software that uses a geometrical optics approximation to model the propagation of acoustic signals in a three-dimensional environment.

The Bellhop simulator uses the ray-tracing technique to model the path of acoustic signals from a transmitter to a receiver through the water medium. The simulator takes into account the environmental conditions, such as water temperature, salinity, pressure, and bottom composition, and generates a 3D model of the underwater environment.

The simulator calculates the transmission loss, arrival times, and amplitudes of the acoustic signals at the receiver, and can provide valuable information for designing and optimizing UWA communication systems. The simulator can also be used to analyze the effects of different system parameters, such as frequency, power, and modulation scheme, on the system performance.

Bellhop simulator is a valuable tool for UWA communication system designers and researchers, as it can provide insights into the complex propagation characteristics of acoustic signals in water. The simulator can help to optimize system performance and reduce the time and cost associated with experimental testing.

CHAPTER 4

BELLHOP SIMULATOR

CHAPTER-4

BELLHOP SIMULATOR

4. BELLHOP

- BELLHOP is a beam tracing model for predicting acoustic pressure fields in ocean environments.
- BELLHOP can produce a variety of useful outputs including transmission loss, eigenrays, arrivals, and received time-series. It also allows for range-dependence in the top and bottom boundaries (altimetry and bathymetry), as well as in the sound speed profile (SSP).
- BELLHOP is implemented in Fortran, Matlab, and Python and used on multiple platforms (Mac, Windows, and Linux).

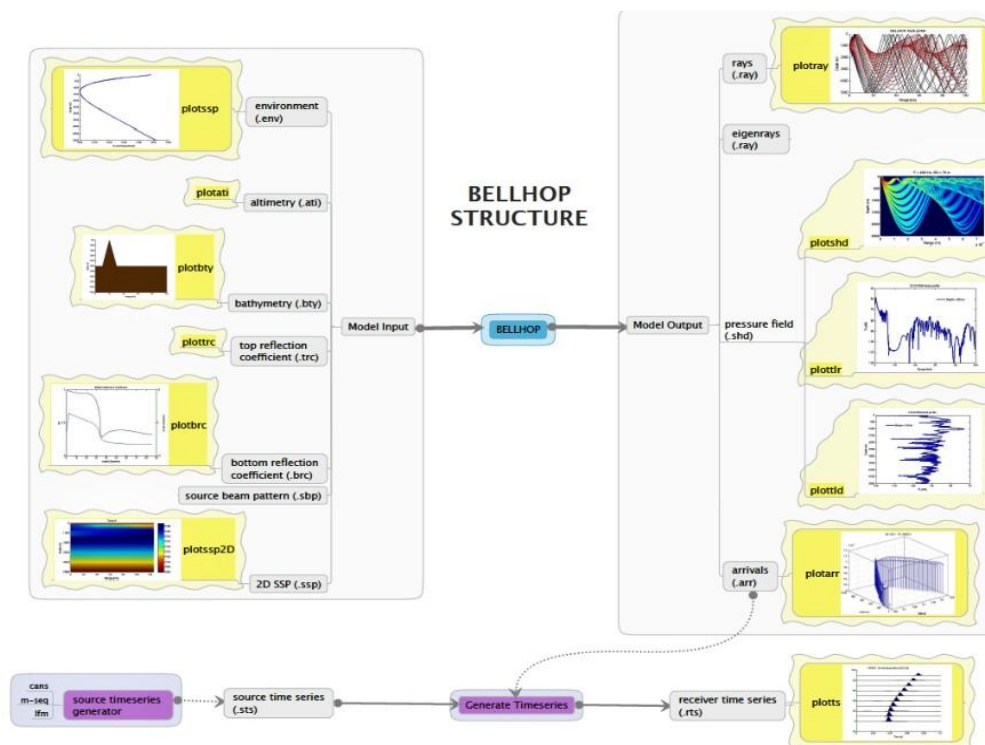


Figure 4: Bellhop

structure

- Underwater communication channel is a relatively difficult transmission medium due to the variability of link quality depending on location and applications.
- Before deploying any kind of vehicles underwater, one should predict the underwater communication system performance which is based on the sound frequency transmitted underwater.

4.1 NEED OF BELLHOP

- Prior to deploying robots, predict communication system performance,
- Provide guidance on best physical layout to deploy underwater vehicles,
- Provide estimates on parameters for link budget calculation,
- Because it will provide you a rough idea about how far you can communicate within network which is also known as an operation range for communication.
- BELLHOP reads these files depending on options selected within the main environmental file.
- There are various options for which you can run bellhop are:
 - Ray tracing option,
 - Eigenray option,
 - Transmission loss option,
 - An arrivals calculation option.

BELLHOP is a highly efficient ray tracing tool, which is designed to perform acoustic ray tracing for a given sound speed profile $c(z)$ or a given sound speed field $c(r, z)$, in ocean waveguides with that or variable absorbing boundaries. Output options include ray coordinates, travel time, amplitude, eigenrays, acoustic pressure or transmission loss (either coherent, incoherent or semi-coherent). Input of BELLHOP is environmental file (.env) to carryout ray tracing and transmission loss, the inputs are given in table 1 that we choose to design. env,

where we have shown the parameters used for modelling two different channels.

It can be used to predict acoustic pressure fields in ocean environments. The beam tracing model is basically described as an approximation of a given source by a fan of beams through the medium.

Then, the interested quantities e.g., acoustic pressure, are computed at a specified location by summing the contribution of each of the individual beams. Bellhop can produce a variety of useful outputs including transmission loss, eigenrays, arrivals and received time-series. It allows for range-dependence on the top and bottom boundaries, and the sound speed profiles as well.

The BELLHOP software determines which function is invoked and what data is produced according to the running parameters contained in the environmental file. Output data include ray coordinates, eigenrays, acoustic pressure, travel time and transmission loss (coherent, incoherent or semi-coherent). The output files can be read and plotted using the BELLHOP functions `plotray`, `plotsd`, `plottlr`, `plottld` and `plotarr`.

4.2 BELLHOP Implementation:

The transmission losses are dependent on selected frequency, thus in order to perform simulation for a small band, BELLHOP consider centre frequency for the configuration, using the arrival time and transmission loss (TL) of centre frequency the behaviour of complete small band can be estimated.

Using the same trend we divided our large 9k to 90k Hz frequency band into four small bands namely Extended Low frequency band (9 to 16 kHz), Low frequency band (16 to 32 kHz), Mid frequency band (31 to 53 kHz) and high frequency band (50 to 90 kHz) to obtain the complete behaviour of overall band using two different channel environments. These frequencies are then provided to BELLHOP tool to calculate the transmission losses and ray tracing.

To obtain the transmission losses results for extended low frequency band (9k to 16k Hz), we selected the centre frequency.

4.3 Bellhop Acoustic Toolbox

AcTUP was initially developed as the “Acoustics Toolbox Front-End” by Alec Duncan to facilitate the rapid application of, and performance comparisons between, the different acoustic propagation codes available in Mike Porter’s Acoustic Toolbox (AT).

This was achieved by providing the user with a unified environment model, automating the creation of code-specific initialization files and by providing intelligent guesses and select.

The development of AcTUP from version 1.6 to the current version was largely funded by the defence Science & Technology Organization (DSTO). ion guidelines for code-specific parameters.

AcTUP is research level code and a little clunky at times but the menu driven format makes it mostly self-explanatory.

4.3.1 Installing and Launching AcTUP

- AcTUP v2.2L requires only MATLAB version 6 or later. In addition to this the full release specification requires the MATLAB Signal Processing Toolbox Note that version 2.2 was developed using MATLAB R2000a.
- All the files you require for installing AcTUP v2.2L are available from the Centre for Marine Science and Technology Website @

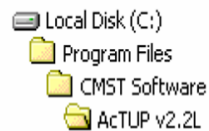
<http://www.cmst.curtin.edu.au/products/actoolbox/index.html>

- The appropriate link will allow you to download the following self-extracting zip file:
- AcTUP v2.2 lite.exe

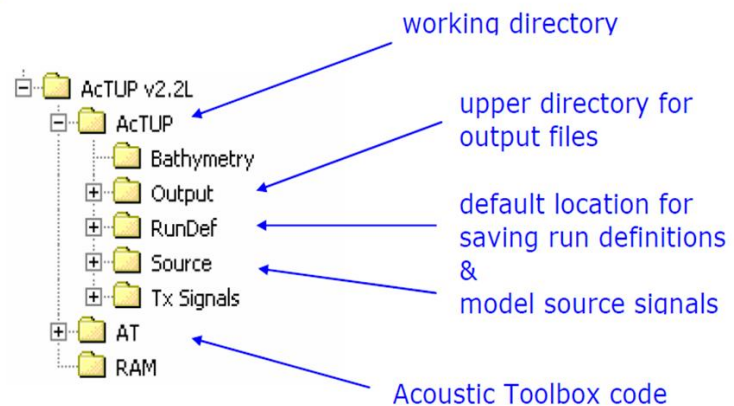
4.3.2 Installing Files

To install AcTUP v2.2(L) simply run the executable above

After extract the topmost AcTUP directory will locate directly below the default target directory as shown in the figure below:



Below the target directory, your directory structure (no matter what target was selected) will look like:



You should AVOID editing or deleting installed directory names.

4.3.3 Setting Up AcTUP

- You only need to complete the following instructions if you DID NOT use the default target directory in the installation process. if you used some other target directory.
- Navigate to the folder

C:\Program Files\CMST Software\AcTUP v2.2L\AcTUP\Source
- Open Get AcDirectoryInfo.m

4.3.4 Running AcTUP

- In order to run AcTUP you must point MATLAB at the working directory AcTUP.
- You must then run the top level m file actup.m. The example below is for the default installation case.
- **>> cd('C:\Program Files\CMST Software\AcTUP v2.2L\AcTUP');**
- **>> actup**
- Edit the line ...

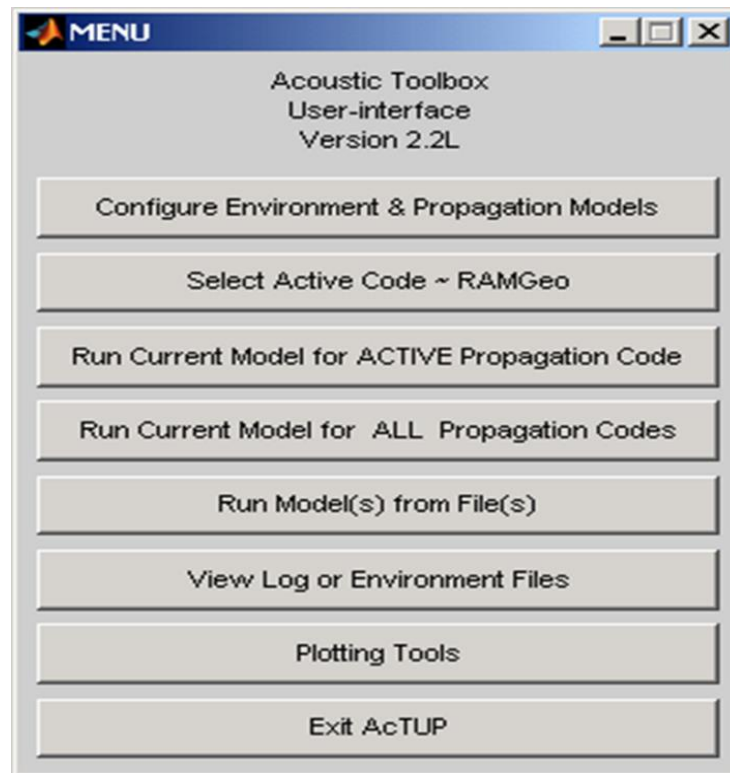
Top = 'C:\Program Files\CMST Software\AcTUP v2.2L\';

... to reflect the correct location of the AcTUP v2.2L directory. Save and close this file. Now point MATLAB at the working directory by

4.3.5 Summarizing bellhop simulator

- Main Menu
- Configure Environment & Propagation models
- Environment & Propagation Definitions
- Edit code-dependent propagation parameters
- Edit code-independent propagation parameters
- Plotting Tools

4.3.6 Main Menu



4.3.7 Configure Environment & Propagation models:

Launches another menu which allows the user to define the model environment, set basic propagation parameters, to set parameters specific to the available propagation codes, to save all environment and propagation model parameters as a single Run Definition, and to load previously saved Run Definitions.

4.3.8 Select Active Code ~ <code>:

Selects which of the available codes is used in modeling propagation when the button below this one is selected.

4.3.9 Run Current Model for Active Propagation Code:

Runs the environment/propagation model currently in memory using the selected propagation code. Note ActUP does not check whether the code-specific parameters have been set for the selected propagation code.

4.3.10 Run Current Model for ALL Propagation codes:

Runs the environment/propagation model currently in memory using all available propagation codes. Note that in this mode Bounce and Fields are only run in conjunction with Bellhop and Scooter respectively.

4.3.11 Run Model(s) from File(s):

This options enables batch mode processing. The user is prompted to select a directory containing one or more Run Definitions. ActUP will sequentially load and run each Run Definition file.

4.3.12 View Log or Environment Files:

Allows user to open and view log (.prn) and environment definition (.env) files. The log files are text files which contain a log of the command line instructions issued by ActUP and any errors generated by them.

4.3.13 Environment & Propagation Definitions



4.3.14 Load DEFAULT Run Definition:

The default configuration has preloaded codes specific parameters for each available propagation code, and can be run as a first test after installation.

4.3.15 Load Run Definition:

To date there is no specific file name extension identifying Run Definition objects and instead we use the default.

4.3.16 Save Run Definition:

While you may use a different extension from .mat it is advised that you stick with this format. If you save both input and output files in an area designated for AcTUP modelling then there is no need to distinguish these .mat files from any others since they are the only ones used by AcTUP.

4.3.17 Edit Environment:

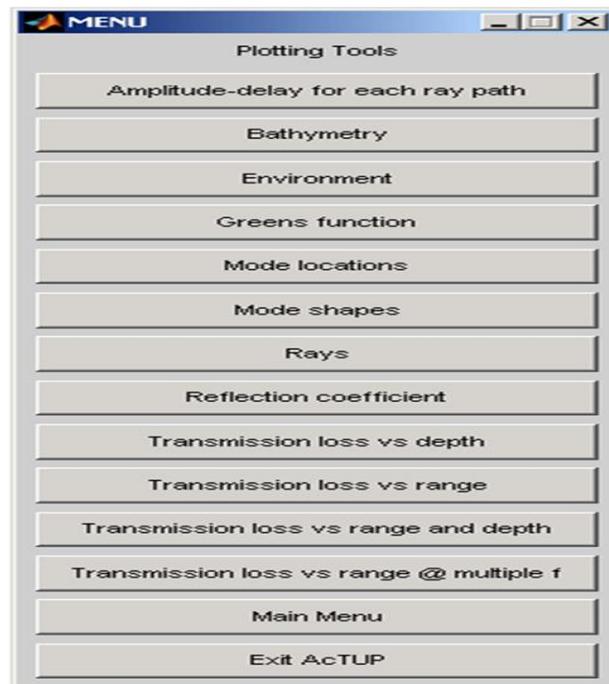
This launches an interface to edit the model environment. This consists of an array of range-independent Environment objects, each defined at a specific range. The range-independent codes (all codes, except RAM* in this version) only use the first element of the Environment array.

Each Environment object is a collection of range independent layers of acoustic media. The first layer is always the water column and the second layer is always the upper-most layer of the substrate. Each layer consists of lists of depth- (z-) dependent properties which must be specified.

4.3.18 Edit Code-Dependent Propagation Parameters:

Information additional to that provided above is required for some codes (i.e. Bellhop, KrakenC, RAMGeo and RAMSGeo). Here the user can set appropriate parameters values for each code. Note that most of the values for RAMGeo and RAMSGeo are common.

4.3.19 Plotting Tools:



- **Amplitude-delay for each ray path:** Plot the data from .arr files output by Bellhop in “A” mode.
- **Bathymetry:** Plots bathymetry data from bathymetry files in the input format required by AcTUP.

ascii format:

row #	format	comments
1	%d	number of points
2 to N+1	%f %f	r [km] z [m]

Table 4. Ascii format

- **Environment:** Plots water and substrate physical and acoustic parameter profiles for the currently loaded environment model.
- **Green’s function:** Plots the results generated by Scooter.
- **Mode locations & Mode shapes:** Uses output from Kraken & KrakenC.

- **Rays:** Plots ray paths generated by Bellhop in “R” mode.
- **Reflection coefficient:** Plot the data from .brc files output by Bounce.
- **All Transmission loss plots:** Plots output from .shd files generated by Bellhop (C, S & I modes), Kraken, KrakenC, Fields, and RAM*.

CHAPTER 5

RESLUTS AND CONCLUSION

CHAPTER-5

5.1 RESULTS

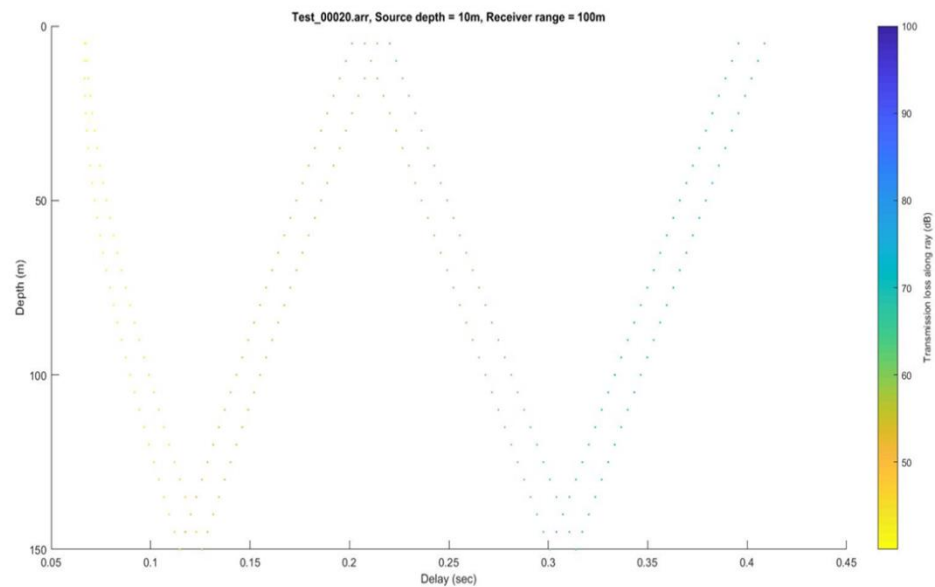


Figure 5. Delay vs Depth

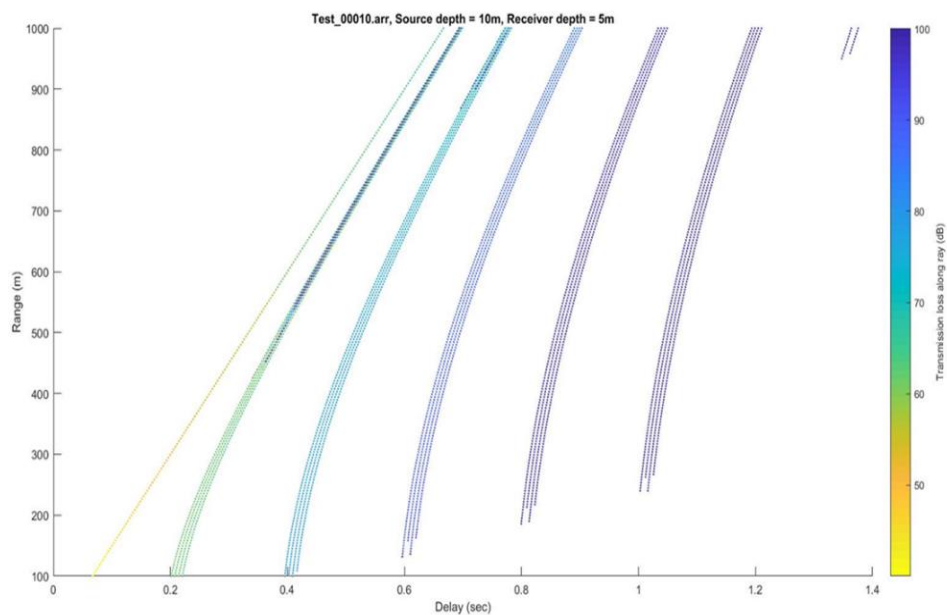


Figure 6. Delay vs Range

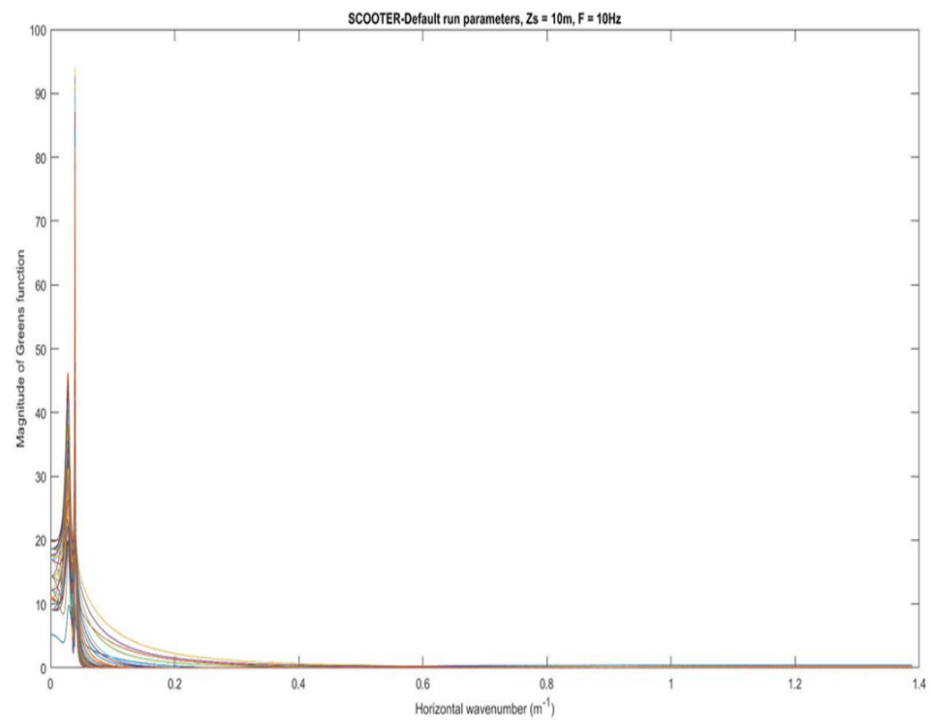


Figure 7. Horizontal wavenumber vs Magnitude of Greens function

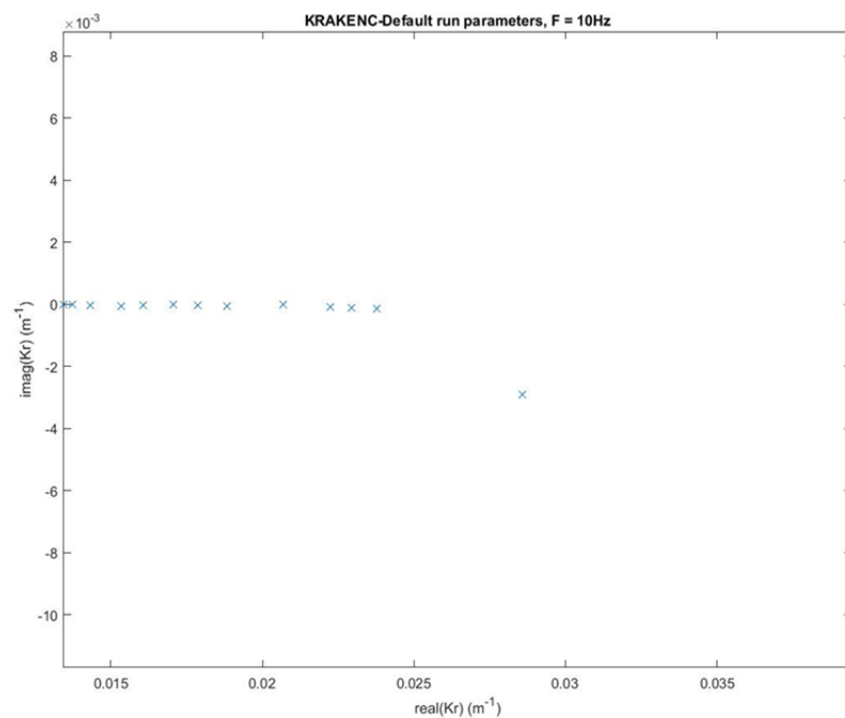


Figure 8. Real vs Imaginary

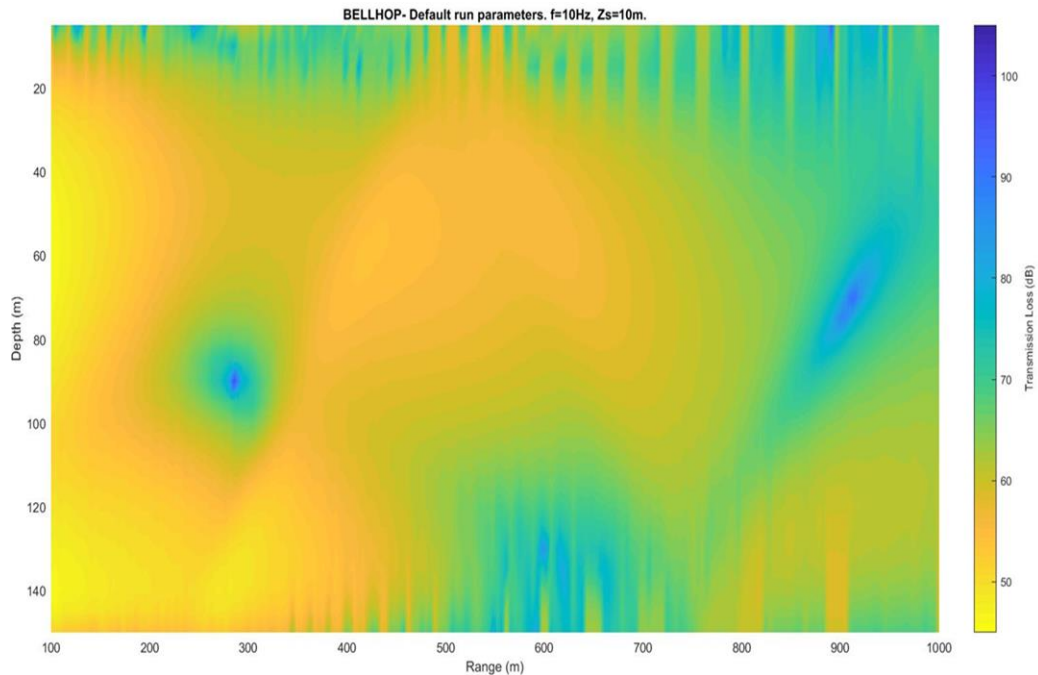


Figure 9. Range vs Depth

5.2 CONCLUSION

The objective was to design a dynamic platform where using different oceanic condition we would be able to examine an acoustic transmission. The analysis work revealed that more efficiency can be achieved by selecting appropriate components for sound propagation in different scenarios, the work can be utilized for variety of applications including masking of platform signatures by deploying false acoustic tones, this study can be the basis for various underwater acoustic analysis work and defense applications.

In conclusion, underwater acoustic communication is a complex process that is dependent on several parameters such as transmission loss, range, depth, and multiple frequencies. The Bellhop ray-tracing software is a useful tool for predicting the transmission loss of sound waves at different frequencies over a given range in underwater environments. The software takes into account factors such as the frequency of the sound waves, the properties of the medium, and the presence of obstacles, to provide accurate predictions of transmission loss in different scenarios. By understanding the relationship between transmission loss, range, depth, and multiple frequencies.

Researchers and engineers can design and optimize underwater acoustic communication systems for a wide range of applications, including underwater exploration, military communication, and environmental monitoring.

5.3 OUTCOMES

Through this project entitled ‘Analysis of UWA Communication Independent Parameters Using Bellhop Simulator’ all Program Outcomes (POs) PO1, PO2, PO3, PO4, PO11, PO12 also Program Specific Outcomes (PSOs) PSO1 are attained.

5.4 FUTURE SCOPE

The future scope of underwater acoustic (UWA) communication on dependent parameters using the Bellhop simulator is significant. With advancements in technology and the increasing demand for underwater communication, the use of the Bellhop simulator is likely to become more widespread and sophisticated.

One potential area for future research is the development of more accurate and efficient algorithms for predicting transmission loss in different underwater environments. This could involve the use of machine learning techniques to analyze large amounts of data and identify patterns in the behavior of sound waves in different scenarios.

Another area of research could focus on the development of new materials and technologies for improving the transmission of sound waves in underwater environments. For example, new materials could be developed that reduce the absorption and scattering of sound waves, or new technologies could be developed for enhancing the efficiency of acoustic transducers and receivers.

In addition, the use of UWA communication systems for environmental monitoring and underwater sensing is likely to increase in the future. This could involve the deployment of large networks of sensors and communication nodes to monitor oceanographic parameters, such as temperature, salinity, and pressure, as well as the presence of marine life and other underwater phenomena.

Overall, future of UWA communication on dependent parameters using the Bellhop simulator is promising, and continued research and development in this area is likely to lead to significant advances in underwater communication and sensing technology.

5.5 REFERENCE

1. J. Peterson and M. Porter, "Ray/Beam Tracing for Modelling the Effects of Ocean and Platform Dynamics", IEEE Journal of Oceanic Engineering, vol. 38, no. 4, pp. 655-665, 2013.
2. Porter M.B. and Bucker H.P. Gaussian beam tracing for computing ocean acoustic fields. J. Acoust. Soc. America, 82(4):1349-1359,1987.
3. Jensen F., Kuperann W., Porter M., and Schmidt H. Computational Ocean Acoustics. ALP Series in Modern Acoustics and Signal Processing, New York, 1994.
4. MacGillivray, "Modelling underwater sound propagation from an airgun array using the parabolic equation method", The Journal of the Acoustical Society of America, vol. 122, no. 5, p. 2942,2007.
5. N. Chotiros and M. Stern, "High-frequency acoustic wave propagation modelling in shallow water", The Journal of the Acoustical Society of America, vol. 100, no. 4, p. 2701, 1996.
6. R. Frosch, "Underwater Sound: Deep-Ocean Propagation: Variations of temperature and pressure have great influence on the propagation of sound in the ocean.", Science, vol. 146, no. 3646, pp. 889-894, 1964.
7. Milica Stojanovic, "Underwater Acoustic Communication Channels: Propagation Models and Statistical Characterization. ", IEEE Communications Magazine, January 2009.
8. M. B. Porter, Ocean Acoustics Library, Bellhop, <http://oalib.hlsresearch.com/>.
9. Porter M.B. The BELLHOP Manual and User's Guide
10. P. H. Milne, Underwater acoustic positioning systems. E. & F.N. Spon Ltd, London, 1983.
11. "Bellhop gaussian beam/finite element beam code," Available in the Acoustics Toolbox, <http://oalib.hlsresearch.com/>lRays, 2015.

12. R. Thiele, "NATO model study," SACLANT Undersea Research Rep., SR-149, 1989.
13. M. B. Porter and E. L. Reiss, "A numerical method for Ocean acoustic normal modes," J. Acoustic. Soc. of Am., vol. 76, no. I, pp. 244-252, 1984.
14. D. Lee, G. Botseas, and I. S. Papadakis, "Finite difference Solutions to the parabolic equation," J. Acoustic. Soc. Am., vol. 70, pp. 795-800, 1981.
15. F. Tappert, "The parabolic equation method," in Wave Propagation und Underwater Acoustics. I. B. Keller and I. S. Papadakis, Eds. New York: Springer-Verlag. 1977.
16. Zhang, S., & Sun, L. (2014). Modeling and Analysis of Underwater Acoustic Communication Channels with Bellhop Simulator. International Journal of Modeling, Simulation, and Scientific Computing, 5(03), 1440008. doi: 10.1142/s1793962314400088
17. Li, M., Liang, L., & Han, B. (2016). Simulation of Underwater Acoustic Communication Channels using Bellhop. International Journal of Distributed Sensor Networks, 12(9), 1550147716671252. doi: 10.1177/1550147716671252
18. Zhang, Z., Zhang, G., & Han, G. (2016). Analysis of Underwater Acoustic Communication Channels using Bellhop. Journal of Marine Science and Engineering, 4(3), 51. doi: 10.3390/jmse4030051
19. Peppas, K., & Frangiadakis, N. (2014). Bellhop-based Simulation of Underwater Acoustic Communication Channels. Journal of Signal and Information Processing, 5(4), 101-108. doi: 10.4236/jsip.2014.54014
20. Cui, H. Y., Chen, Y. B., & Gao, J. L. (2014). Evaluation of Underwater Acoustic Communication Performance using Bellhop Simulation Model. Journal of Electronics & Information Technology, 36(1), 22-28.
21. Li, W., Li, L., & Wang, H. (2018). Analysis of Underwater Acoustic Communication System Performance in Different Water Conditions. International Journal of Distributed Sensor Networks, 14(2), 1550147718756810.

22. Yu, X., Zhang, W., & Liu, X. (2018). Optimization of Underwater Acoustic Communication System Design using Bellhop Simulation Model. *Journal of Physics: Conference Series*, 1013(1), 012045.
23. Zhang, J., Gao, X., & Sun, J. (2018). Analysis of the Effect of Distance Attenuation on Underwater Acoustic Communication System Performance. In *2018 3rd IEEE International Conference on Image, Vision and Computing (ICIVC)* (pp. 764-768). IEEE.
24. Ma, Y., Li, X., & Huang, X. (2019). Simulation Analysis of Underwater Acoustic Communication Performance in Shallow Water. In *2019 IEEE International Conference on Artificial Intelligence and Computer Applications (ICAICA)* (pp. 18-22). IEEE.